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Micro I, class 2

We finished our consideration of common axioms: *completeness*, *transitivity*, *continuity* (sequential and mixture), and *monotonicity* (regular and strong); *local nonsatiation* (LNS), and *convexity* (regular and strict).

And we proved together *this* theorem on a preference representation.

Theorem. *Let $X = \mathbb{R}^L_+$ and suppose that $\succsim \subset X \times X$ is* ***reflexive*** *($x \succsim x$ for every $x \in X$),* ***transitive****,* ***mixture continuous****, and* ***strongly monotone****. Then there is a function $u : X \to \mathbb{R}$ that represents $\succsim$.*

Since any *representable* relation must be complete, the listed axioms *imply* completeness. And since sequential continuity and completeness *imply* mixture continuity, those two axioms could be substituted in the theorem statement. In that case there is a *continuous* representation, the version of the Theorem that MWG present and prove.

With this Theorem (or MWG's version) in hand we can rewrite standard consumer theory as

$$(UM) \qquad \max_{x \in X} u(x) \quad \text{subject to} \quad p \cdot x \leq w.$$

By default we will assume that $X \subseteq \mathbb{R}^L_+$ is closed and that all utility representations are *continuous* and *locally non-satiated.* We will usually assume that the consumption set is $\mathbb{R}^L_+$, but will sometimes consider other possibilities. The demand correspondence, $d(p, w) = \{y \in B(p, w) \mid u(y) \geq u(x) \ \forall x \in B(p, w)\}$ characterizes the dependency of solutions on the price-wealth pair. Continuity of u, $p \gg 0$, and closedness of $X \subseteq \mathbb{R}^L_+$ together imply that a solution always exists, by the Weierstauss theorem—good, we have something to say—and LNS of u implies that, at any solution, the budget inequality must hold as an equality, which I refer to as *budget balance.*

Next class I will begin with two questions: how can we find functional forms to estimate (or calibrate) demand? and how can we derive testable implications from the theory in equation (UM)?

As a preamble, I will start by writing down the *Kuhn-Tucker conditions* for the UM problem. The K-T conditions can be found in my notes on *Concave Programming*, posted to Canvas in the *Notes* folder.